CHAPTER

3

MAN-MADE FIBRES

U62VUX

LEARNING OBJECTIVES

- **To understand the manufacturing process of rayon, acetate and nylon.**
- **To acquire knowledge about the properties and uses of rayon, acetate and nylon.**

3.1 INTRODUCTION:

Man-made fibres are fibres which as the name say are made by man with a combination of natural products along with chemicals. The main advantage of these fibres is properties of the natural product used can be improved. Manmade fibres are made by preparing a viscous solution which is passed through the spinneret to form the fibre, which is cooled and solidified as it comes out of the spinneret. It is a small round metal with large number of tiny holes.

3.2 RAYON:

The rayon fibre is made with pure cellulose. Cellulose is the substance which is obtained from the cell walls of the woody part of trees and cotton plant. Cellulose is commonly used for making products such as paper. The rayon fibre that has been formed with regenerated or re-formed cellulose substance is called as regenerated cellulose fibres.

Rayon is the first man-made fibre and its production has been prophesied as long ago as 1664 by Robert Hooke, an English naturalist. He believed that it was possible to make an "artificial glutinous" composition which can resemble the silkworm fibre. Man-made textile fibres were officially recognized in 1925 when the Federal Trade Commission (FTC) permitted the use of the name "rayon" from yarns obtained from cellulose or its derivatives. With the increase in production of man-made fibres, FTC ruled again in 1937 that any fibre or yarn produced chemically from cellulose must be designated as rayon.

TDD_Final_Chapter03.indd 23 21-02-2019 11.11.59 AM